COURSE SLIDES

# Certified Lean Six Sigma Green Belt (CLSSGB) Training

WSQ Course Code: TGS-2025055775

Conducted by Tertiary Infotech Academy Pte Ltd | UEN 201200696W

Version 1

ADMIN

# Copyright

YOUR TRAINER - GENERAL

# About the Trainer

?

## Your Trainer

General Trainer template -
to be completed by the trainer

**NAME**

**TITLE / DESIGNATION**

**QUALIFICATIONS**

**AREAS OF EXPERTISE**

**TRAINING & INDUSTRY EXPERIENCE**

**CONTACT**

YOUR TRAINER

# About the Trainer

AA

## Dr. Alfred Ang

Principal Trainer
Tertiary Infotech Academy Pte. Ltd.

**ROLE**

Principal Trainer, Tertiary Infotech Academy Pte. Ltd.

**QUALIFICATIONS**

PhD - specialises in AI, automation and software engineering.

**DELIVERS**

WSQ courses on Lean Six Sigma, AI automation and app development.

**FOUNDER**

Founder and lead instructor at Tertiary Infotech / Tertiary Courses.

ADMIN

# Course Agenda

- Day 1: Lean Six Sigma basics, project selection, SIPOC, process mapping, facilitation, A3, KPIs and baseline charts.
- Day 2: Basic statistics, capability, root cause analysis, FMEA, countermeasures, control plan and project close.
- Final session: A3 presentation readiness, assessment briefing and course wrap-up.

HOUSEKEEPING

# Ground Rules

1. Set your mobile phone to silent mode.
2. Participate actively - no question is too small.
3. Mutual respect: agree to disagree.
4. One conversation at a time.
5. Be punctual; return from breaks on time.
6. 75% attendance is required.

# Learning Outcomes

- Explain Lean, Six Sigma, DMAIC, COPQ, Kano and Green Belt responsibilities.
- Select and scope a suitable improvement project.
- Create SIPOC, process maps, stakeholder analysis, Kaizen plan and A3 report.
- Define KPIs, collect baseline data and build run/Pareto charts.
- Apply basic statistics, root cause analysis, FMEA and countermeasures.
- Create a control plan, response plan and project closeout presentation.

ADMIN

# Digital Attendance and LMS

- Take AM, PM and assessment digital attendance when instructed by the trainer.
- Minimum attendance requirements apply for SSG-funded courses.
- Courseware and assessment instructions are provided through the LMS.
- Submit required assessment evidence before leaving the classroom.

ASSESSMENT

# Assessment Flow Reminder

1. TRAQOM digital attendance
>
2. Assessment digital attendance
>
3. Sit WA then case study
>
4. Submit answers on the LMS
>
5. Sign Assessment Summary Record

The TRAQOM QR code, LMS submission and assessment evidence are handled through the LMS/TMS portal.

ADMIN

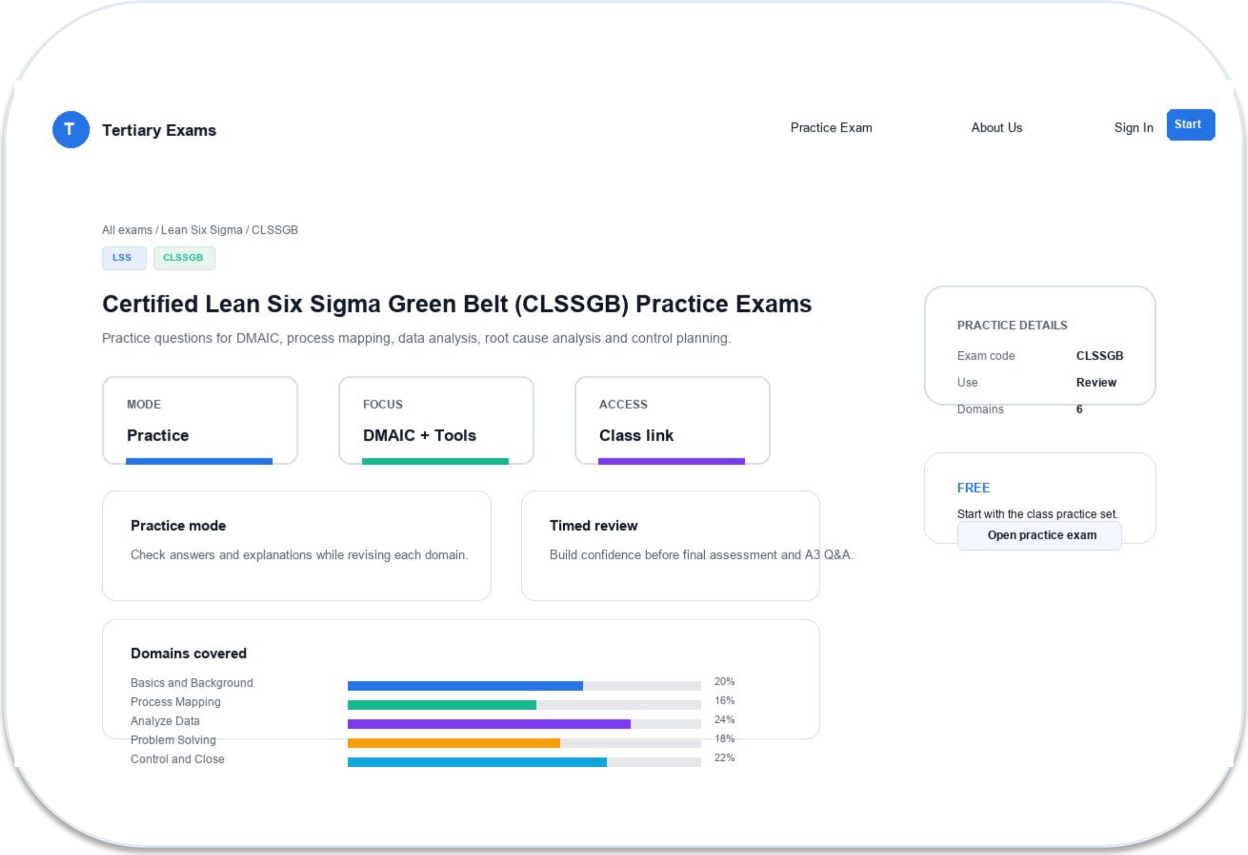
# Practice Project File

- Project selection worksheet, problem statement and goal statement.
- SIPOC, process map, stakeholder analysis, Kaizen plan and A3 report.
- Data collection plan, baseline data, run chart, Pareto chart and statistics worksheet.
- Root cause worksheet, FMEA, countermeasure plan, standard work and control plan.

PRACTICE EXAM

# Take the Online CLSSGB Practice Exams

- Reinforce all six CLSSGB domains with realistic exam-style questions before assessment day.
- Use practice sets to test DMAIC, process mapping, data analysis, FMEA and control planning.
- Review mode helps you check the correct answer and revisit the concept after each question.
- Complete the practice questions from Google Classroom, then bring difficult items to the review session.




**Access and practise at:**
**Google Classroom > Practice Exams**

# DMAIC Project Flow

Each lab contributes one artifact to the final Green Belt A3 project story.

**Define**
Problem, customer, scope

**Measure**
Baseline and KPI data

**Analyze**
Patterns and root causes

**Improve**
Countermeasures and pilots

**Control**
Sustain and hand over

KEY CONCEPT

# Lean and Six Sigma

## Lean focus

- Remove non-value-added work
- Reduce waiting, rework, handoffs and motion
- Improve flow and visual management

## Six Sigma focus

- Reduce variation and defects
- Use data to understand performance
- Protect the customer-critical output

TOOLKIT

# Core Green Belt Tools

## Project and process tools

- Project charter and selection matrix
- SIPOC and current-state process map
- Stakeholder analysis and facilitation plan
- A3 report and Kaizen plan

## Data and control tools

- Operational definitions and KPI plan
- Run chart, Pareto chart and basic statistics
- Fishbone, 5 Whys and FMEA
- Control plan and response plan

DOMAIN 1

# Basics and Background

Lean, Six Sigma, DMAIC, Green Belt role, COPQ, Kano and project selection

# Lab 01: Lean Six Sigma Basics and Project Selection

You have been asked to support an improvement initiative. Before mapping or measuring the process, you must choose a project that is valuable, realistic, and appropriate for a Green Belt.

## Objectives

- Explain Lean Six Sigma fundamentals.
- Describe the Green Belt role.
- Identify customer value and COPQ.
- Select a suitable improvement project.

## Deliverables

- Project selection worksheet.
- COPQ notes.
- Kano notes.
- Draft problem and goal statements.

STEP-BY-STEP PRACTICE

# Lab 01 Steps (1)

1. Write your own definition of Lean.
2. Write your own definition of Six Sigma.
3. List the five DMAIC phases and one activity in each phase.
4. Describe what a Green Belt typically does in an improvement project.
5. Select three possible process problems from your workplace or a trainer scenario.
6. For each problem, identify the customer affected.

STEP-BY-STEP PRACTICE

# Lab 01 Steps (2)

7 Estimate cost of poor quality, including rework, delay, errors, complaints, or morale impact.

8 Use the Kano model to classify customer needs as basic, performance, or delight factors.

9 Write Y=f(x) for each project idea.

10 Score each project by impact, feasibility, data availability, and scope.

11 Select the best project for the remaining labs.

12 Draft a problem statement and goal statement.

CHECKPOINT

# Lab 01 Validation

## Validation checks

- The selected project is small enough for a Green Belt.
- The problem statement uses observable facts.
- The goal statement is measurable.
- Customer value is clearly described.

## Review questions

- How do Lean and Six Sigma complement each other?
- What makes a project suitable for a Green Belt?
- Why should COPQ include hidden costs?
- What does Y=f(x) mean in process improvement?

DOMAIN 2

# Process Mapping

SIPOC, current-state mapping, value-added analysis and eight wastes

# Lab 02: SIPOC, Process Map, and Eight Wastes

The selected project needs a shared view of the process. You will map the process and identify where waste may be harming customer value.

## Objectives

- Build a SIPOC.
- Create a current-state process map.
- Identify value-added and non-value-added steps.
- Locate the eight wastes.

## Deliverables

- SIPOC.
- Current-state process map.
- Waste analysis table.
- Measurement point notes.

STEP-BY-STEP PRACTICE

# Lab 02 Steps (1)

1. Define the start and end points of the process.
2. List the major outputs and customers.
3. List the key inputs and suppliers.
4. Create a SIPOC table.
5. Draw a current-state process map.
6. Include decision points, approvals, handoffs, queues, and rework loops.

STEP-BY-STEP PRACTICE

# Lab 02 Steps (2)

7 Label each process step as value-added, business-value-added, or non-value-added.

8 Identify the eight wastes:

9 Mark the two most visible waste areas on the map.

10 Decide where data should be collected in Lab 04.

CHECKPOINT

# Lab 02 Validation

## Validation checks

- SIPOC boundaries match the project.
- Process map shows the real current state.
- Waste examples are tied to actual process steps.
- Measurement points are identified.

## Review questions

- Why should mapping show the current state instead of the ideal state?
- Which wastes are common in office or service processes?
- How can handoffs create delay?
- Why is underused talent considered waste?

DOMAIN 3

# Facilitation

Stakeholders, facilitation behaviours, Kaizen planning and A3 thinking

# Lab 03: Facilitation, Stakeholders, Kaizen, and A3

Green Belts often improve processes by working through people. You need to bring the right people together, keep discussion structured, and document the improvement story clearly.

## Objectives

- Identify stakeholders.
- Plan a facilitated improvement meeting.
- Design a Kaizen event.
- Start an A3 report.

## Deliverables

- Stakeholder analysis.
- Facilitation agenda.
- Kaizen plan.
- Draft A3 report.

STEP-BY-STEP PRACTICE

# Lab 03 Steps (1)

1. List all stakeholders affected by the process problem.
2. Rate each stakeholder by interest and influence.
3. Identify likely supporters, blockers, and subject matter experts.
4. Write one engagement action for each key stakeholder.
5. Create an agenda for a 60-minute improvement meeting.
6. Include objective, ground rules, roles, timing, and expected outputs.

STEP-BY-STEP PRACTICE

# Lab 03 Steps (2)

7 Plan a Kaizen event for one process pain point.

8 Define pre-work, event activities, and follow-up actions.

9 Create an A3 report with background, current condition, goal, and next step sections.

10 Practise explaining the A3 in three minutes.

11 Update the A3 based on peer or trainer feedback.

CHECKPOINT

# Lab 03 Validation

## Validation checks

- Stakeholder actions match stakeholder concerns.
- Meeting agenda has clear outputs.
- Kaizen scope is small and practical.
- A3 report tells the project story simply.

## Review questions

- Why is facilitation important for Green Belts?
- What makes a Kaizen event focused?
- How does an A3 report support communication?
- Why should blockers be engaged early?

DOMAIN 4

# Analyze Data

KPI definitions, data collection, run charts, Pareto, basic statistics and capability

# Lab 04: Data Collection, KPIs, Run Charts, and Pareto

The project team needs baseline data. You will define what to measure, collect or use sample data, and visualize the problem.

## Objectives

- Define a KPI.
- Create a data collection plan.
- Build a run chart.
- Build a Pareto chart.

## Deliverables

- KPI definition.
- Data collection plan.
- Run chart.
- Pareto chart.
- Focus area statement.

STEP-BY-STEP PRACTICE

# Lab 04 Steps (1)

1. Select one primary KPI for the project.
2. Write an operational definition for the KPI.
3. Define unit, defect, opportunity, and target if applicable.
4. Identify data source, owner, frequency, and sample size.
5. Create a data collection plan.
6. Collect baseline data or use a trainer-provided dataset.

STEP-BY-STEP PRACTICE

# Lab 04 Steps (2)

7 Check for missing values, duplicate records, and inconsistent categories.

8 Create a run chart showing performance over time.

9 Add notes for known events or process changes.

10 Create a Pareto chart for defect categories, delay reasons, or error types.

11 Identify the top category contributing to the problem.

12 Decide what issue should move into root cause analysis.

CHECKPOINT

# Lab 04 Validation

## Validation checks

- KPI definition is clear enough for two people to measure consistently.
- Run chart uses time order.
- Pareto chart is sorted from largest to smallest.
- Analysis identifies where to focus next.

## Review questions

- Why do operational definitions matter?
- What does a run chart show that a simple average hides?
- Why is Pareto analysis useful?
- What data quality checks should be done before analysis?

# Lab 05: Basic Statistics and Process Capability

After visualizing the data, you need to summarize baseline performance. The goal is to describe both the center and spread of the process.

## Objectives

- Calculate basic statistics.
- Interpret variation.
- Calculate common Six Sigma metrics.
- Summarize baseline performance.

## Deliverables

- Basic statistics table.
- Histogram if applicable.
- Yield, DPU, DPMO, or sigma worksheet.
- Baseline performance summary.

STEP-BY-STEP PRACTICE

# Lab 05 Steps (1)

1. Open the baseline dataset from Lab 04.
2. Calculate count, mean, median, minimum, maximum, range, and standard deviation.
3. Create a histogram for continuous data if applicable.
4. Identify whether the process has obvious outliers.
5. Calculate yield if good units and total units are known.
6. Calculate DPU if defects and units are known.

STEP-BY-STEP PRACTICE

# Lab 05 Steps (2)

7 Calculate DPMO if opportunities are known.

8 Estimate sigma level using a trainer-provided conversion table if available.

9 Compare actual performance with customer or business target.

10 Write a short baseline summary using numbers and plain language.

11 Identify one question that still needs investigation.

CHECKPOINT

# Lab 05 Validation

## Validation checks

- Statistics are calculated from the correct dataset.
- Variation is discussed, not only the average.
- Defect metrics use correct denominators.
- Baseline summary is clear to a non-statistical reader.

## Review questions

- Why can averages be misleading?
- What is the difference between defects and defective units?
- Why does DPMO include opportunities?
- How can variation affect customer experience?

DOMAIN 5

# Problem Solving

Fishbone, 5 Whys, evidence checks, FMEA and risk prioritisation

# Lab 06: Root Cause Analysis and FMEA

The team now understands the baseline problem. You will identify possible causes and use FMEA to decide where countermeasures are most needed.

## Objectives

- Use Fishbone and 5 Whys.
- Validate suspected causes with evidence.
- Build an FMEA.
- Prioritize process risks.

## Deliverables

- Fishbone diagram.
- 5 Whys worksheet.
- Evidence notes.
- FMEA.
- Prioritized root cause and risk list.

STEP-BY-STEP PRACTICE

# Lab 06 Steps (1)

1. Select the biggest issue from Lab 04 or Lab 05.
2. Create a fishbone diagram.
3. Use categories such as people, process, material, equipment, measurement, environment, and management.
4. Choose two likely causes from the fishbone.
5. Complete 5 Whys for each selected cause.
6. Identify what evidence would support or reject each cause.

STEP-BY-STEP PRACTICE

# Lab 06 Steps (2)

7 Review available data or collect quick observations.

8 Mark each cause as supported, unsupported, or needing more data.

9 Create an FMEA for the process step most related to the problem.

10 List failure mode, effect, cause, and current controls.

11 Score severity, occurrence, and detection.

12 Calculate RPN.

STEP-BY-STEP PRACTICE

# Lab 06 Steps (3)

13 Identify the top risks for countermeasures.

CHECKPOINT

# Lab 06 Validation

## Validation checks

- Root causes are not based only on opinion.
- 5 Whys reaches process-level causes.
- FMEA scores are discussed by the team.
- High-priority risks are clear.

## Review questions

- Why can 5 Whys fail if the team stops too early?
- What is the purpose of FMEA?
- Why should FMEA include current controls?
- How does RPN help prioritize action?

DOMAIN 6

# Countermeasures

Countermeasure selection, 5S, Kanban, poka-yoke and standard work

# Lab 07: Countermeasures, 5S, Kanban, and Poka-Yoke

The team is ready to improve the process. The countermeasures must address validated causes and be simple enough to test quickly.

## Objectives

- Select countermeasures linked to root causes.
- Apply 5S and visual management.
- Use Kanban concepts.
- Design mistake proofing.

## Deliverables

- Countermeasure list.
- Impact-effort matrix.
- 5S action list.
- Kanban or visual control sketch.
- Poka-yoke idea.
- Draft standard work.

STEP-BY-STEP PRACTICE

# Lab 07 Steps (1)

1. List supported root causes and high-priority FMEA risks.
2. Brainstorm possible countermeasures.
3. Remove ideas that do not address a validated cause.
4. Use an impact-effort matrix to prioritize options.
5. Select one or two countermeasures for a small test.
6. Apply 5S to the relevant physical or information workspace.

STEP-BY-STEP PRACTICE

# Lab 07 Steps (2)

7 Identify where visual management would help the team see status or errors.

8 Identify where a Kanban signal could prevent waiting, shortage, or overload.

9 Design one poka-yoke idea to prevent or detect an error earlier.

10 Draft standard work for the improved step.

11 Define test duration, success measure, and rollback condition.

12 Record expected benefits and possible side effects.

CHECKPOINT

# Lab 07 Validation

## Validation checks

- Each countermeasure links to a supported cause or risk.
- Impact-effort matrix supports the selection.
- Standard work is clear enough for another person to follow.
- Test criteria are measurable.

## Review questions

- Why should countermeasures be linked to root causes?
- How does 5S improve process stability?
- What is the purpose of Kanban?
- Why is mistake proofing better than end-of-process inspection?

DOMAIN 7

# Control and Close

Control plan, response plan, handover, A3 closeout and project presentation

# Lab 08: Control Plan, Standard Work, and Project Close

The improvement has been selected and tested. You must make sure the process owner can sustain the result after the project closes.

## Objectives

- Build a control plan.
- Define response actions.
- Finalize standard work.
- Present the project using A3.

## Deliverables

- Control plan.
- Response plan.
- Final standard work.
- Final A3 report.
- Project presentation notes.

STEP-BY-STEP PRACTICE

# Lab 08 Steps (1)

1. List the improved process steps that require control.
2. Identify the KPI or CTQ to monitor.
3. Define target, measurement method, frequency, and owner.
4. Write a response plan for performance outside the expected range.
5. Finalize standard work instructions.
6. Define training actions for team members.

STEP-BY-STEP PRACTICE

# Lab 08 Steps (2)

7. Define audit or check-in frequency.
8. Compare before and after performance if pilot data is available.
9. Update the A3 report with root cause, countermeasure, result, and follow-up.
10. Prepare a five-minute presentation.
11. Present the project story to a peer or trainer.
12. Record lessons learned and remaining risks.

CHECKPOINT

# Lab 08 Validation

## Validation checks

- Control plan has clear owner and frequency.
- Response plan says what to do when performance changes.
- A3 report tells the complete improvement story.
- Handover actions are practical.

## Review questions

- Why do improvements fade after project closure?
- What should a response plan include?
- How does standard work support consistency?
- Why is A3 useful for project communication?

# Final A3 Project Story

- Define the problem and customer impact.
- Measure current performance using agreed KPIs.
- Analyze the largest contributors and validated root causes.
- Improve with countermeasures linked to causes and risks.
- Control the process with ownership, monitoring and response actions.

COURSE CLOSE

# Thank You

- Complete TRAQOM and any required LMS submissions.
- Keep your project artifacts for workplace application.
- Continue improving with data, discipline and respect for the process.